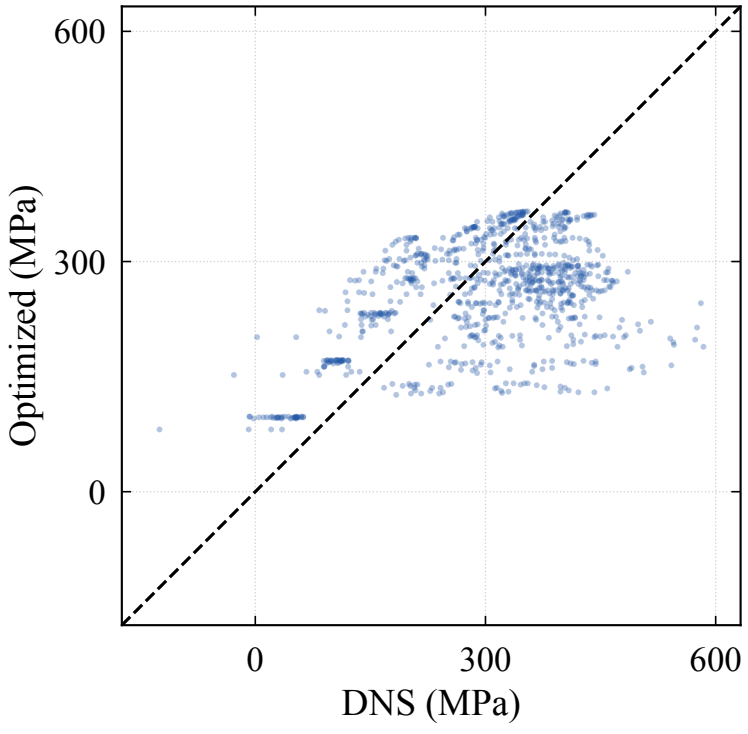
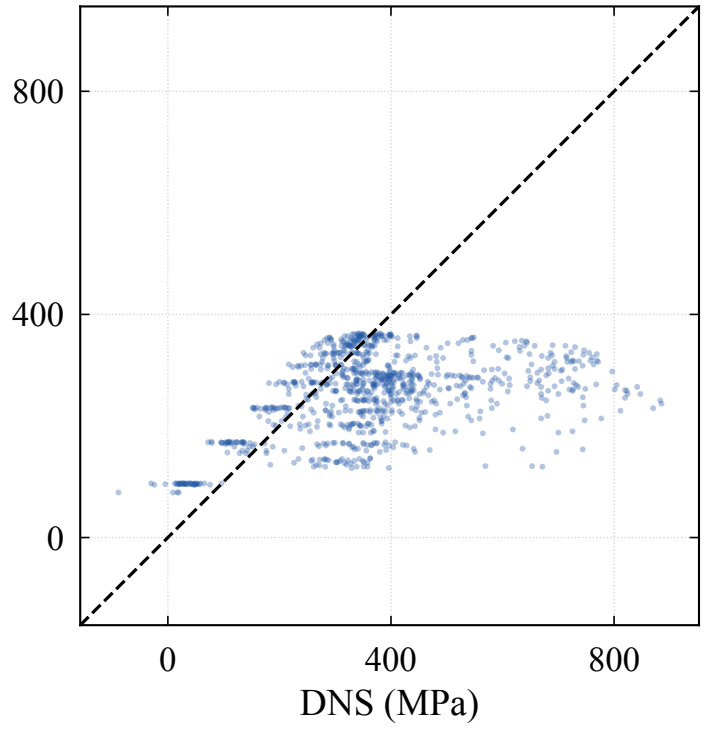
DNS vs optimized: paired-field agreement

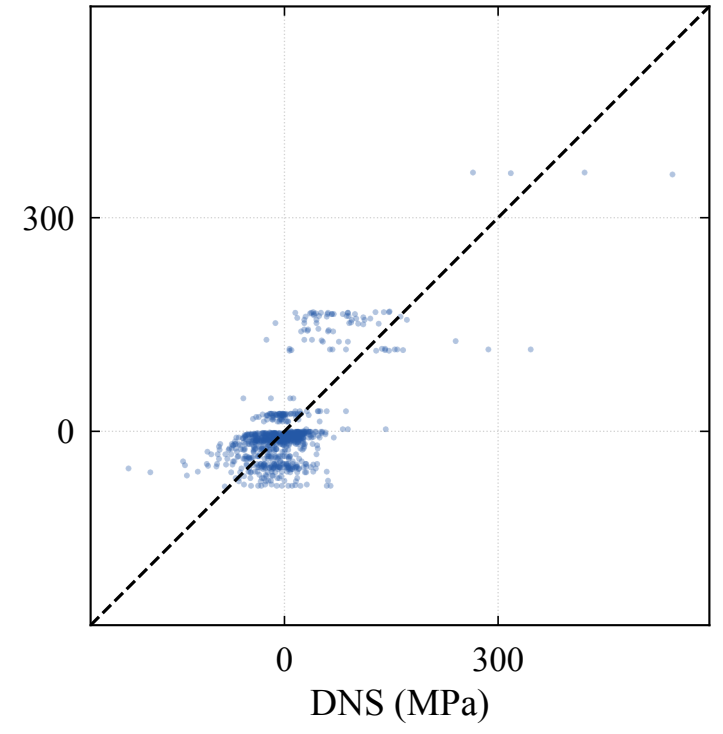
eL2 = 35.3% | R² = 0.04
RMSE = 114 MPa



eL2 = 46.9% | R² = -0.29
RMSE = 192 MPa

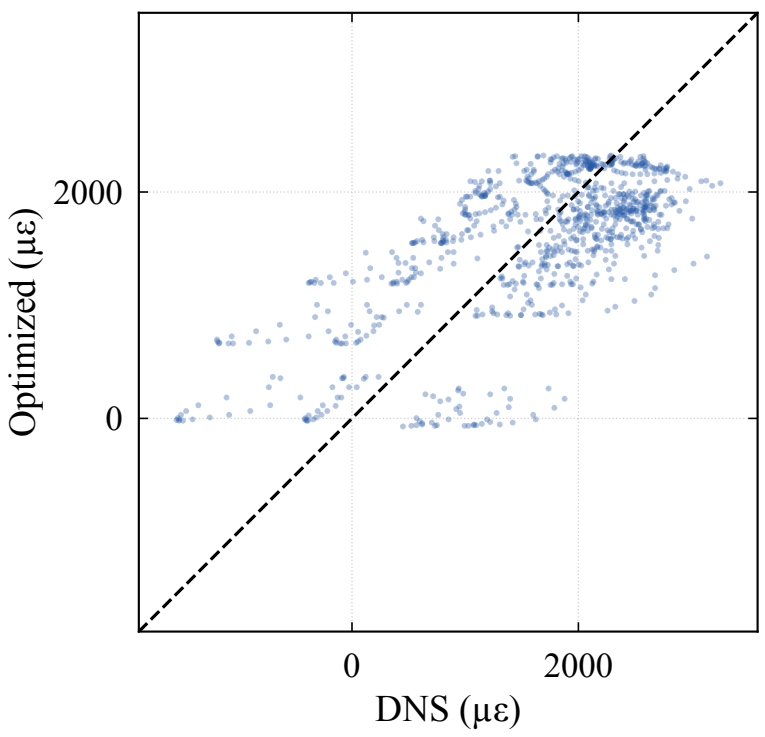


eL2 = 84.5% | R² = 0.29
RMSE = 44.3 MPa



(a) Stress xx

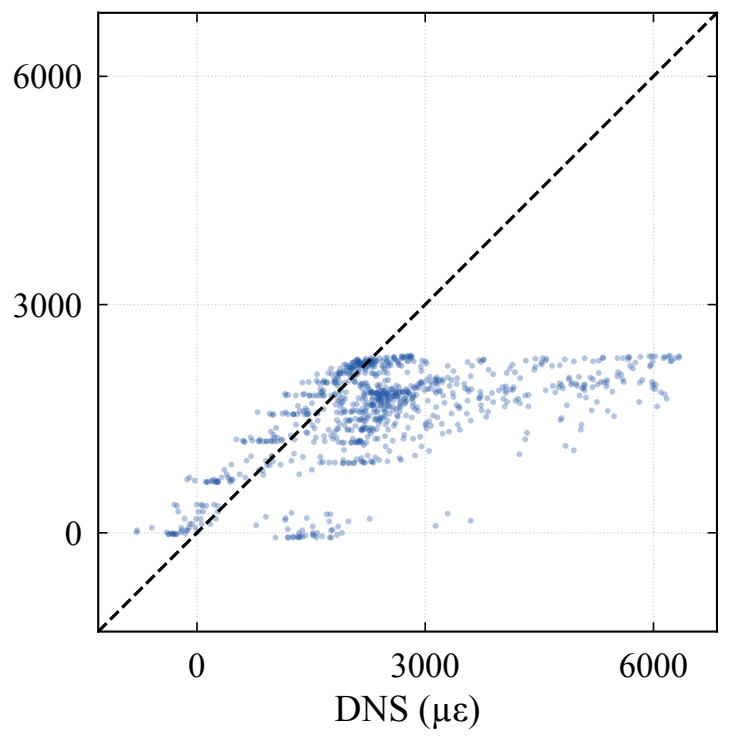
eL2 = 36.6% | R² = 0.46
RMSE = 691 με



(d) Elastic strain xx

(b) Stress yy

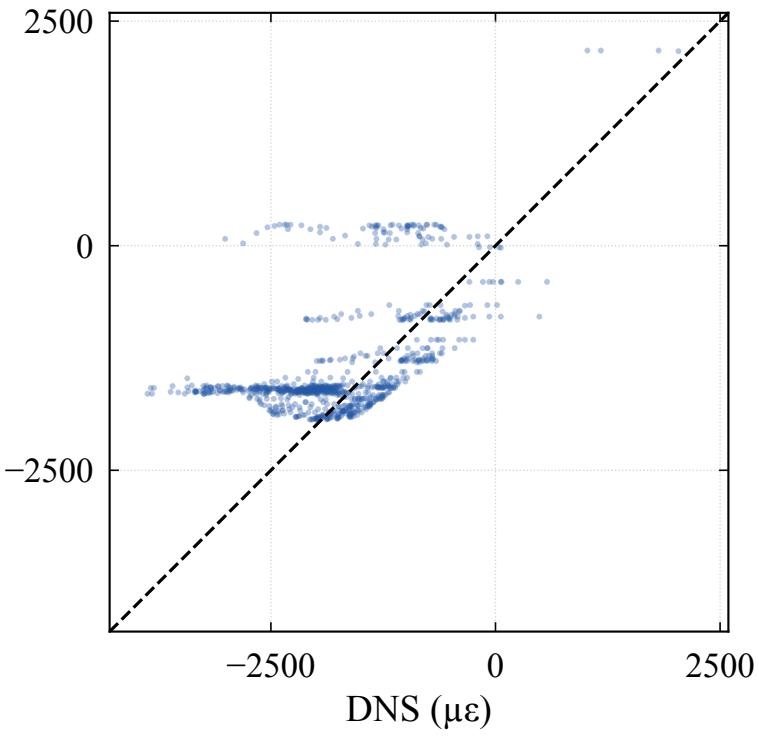
eL2 = 51.4% | R² = -0.10
RMSE = 1.46e+03 με



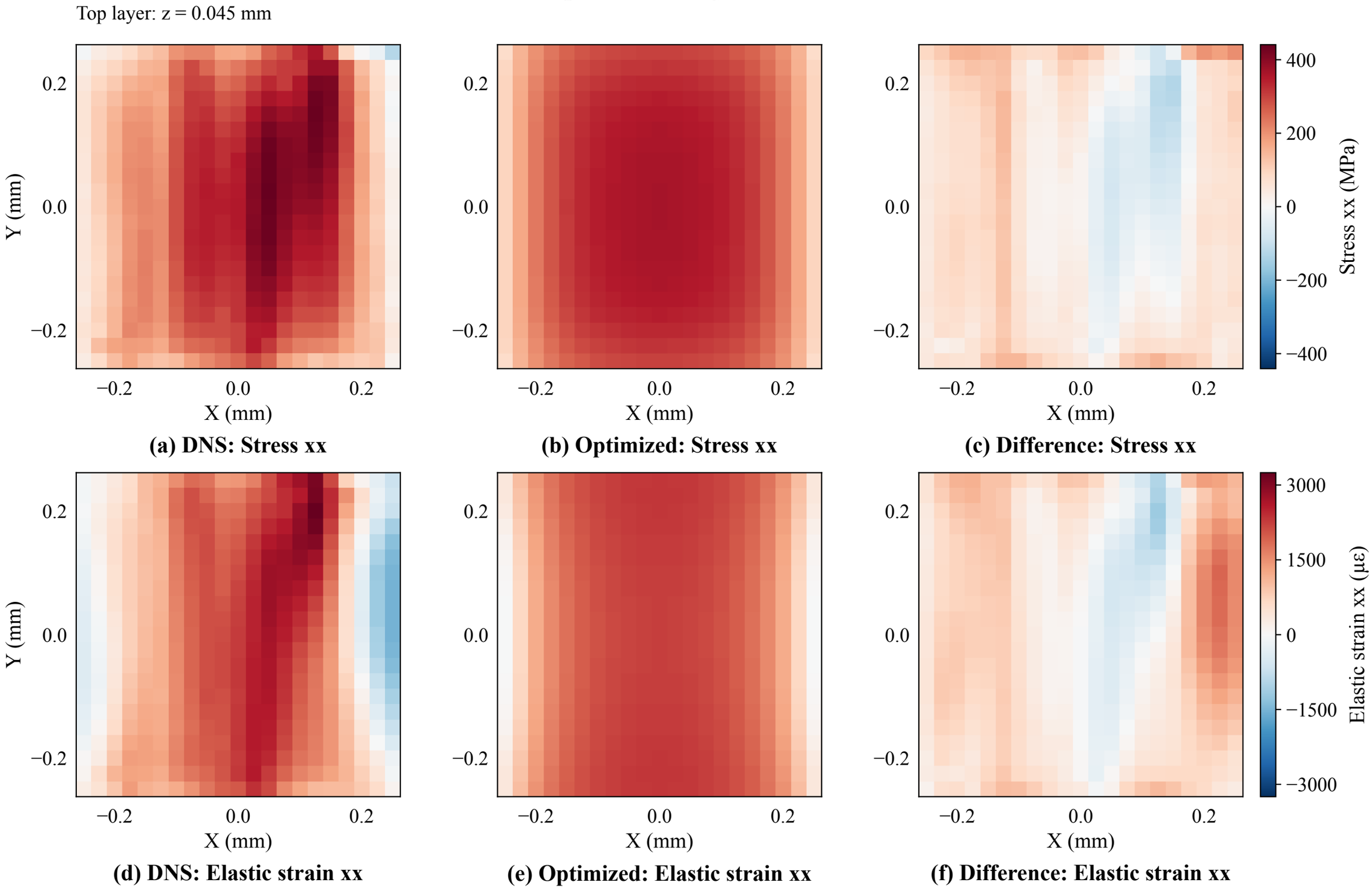
(e) Elastic strain yy

(c) Stress zz

eL2 = 40.9% | R² = -0.05
RMSE = 786 με

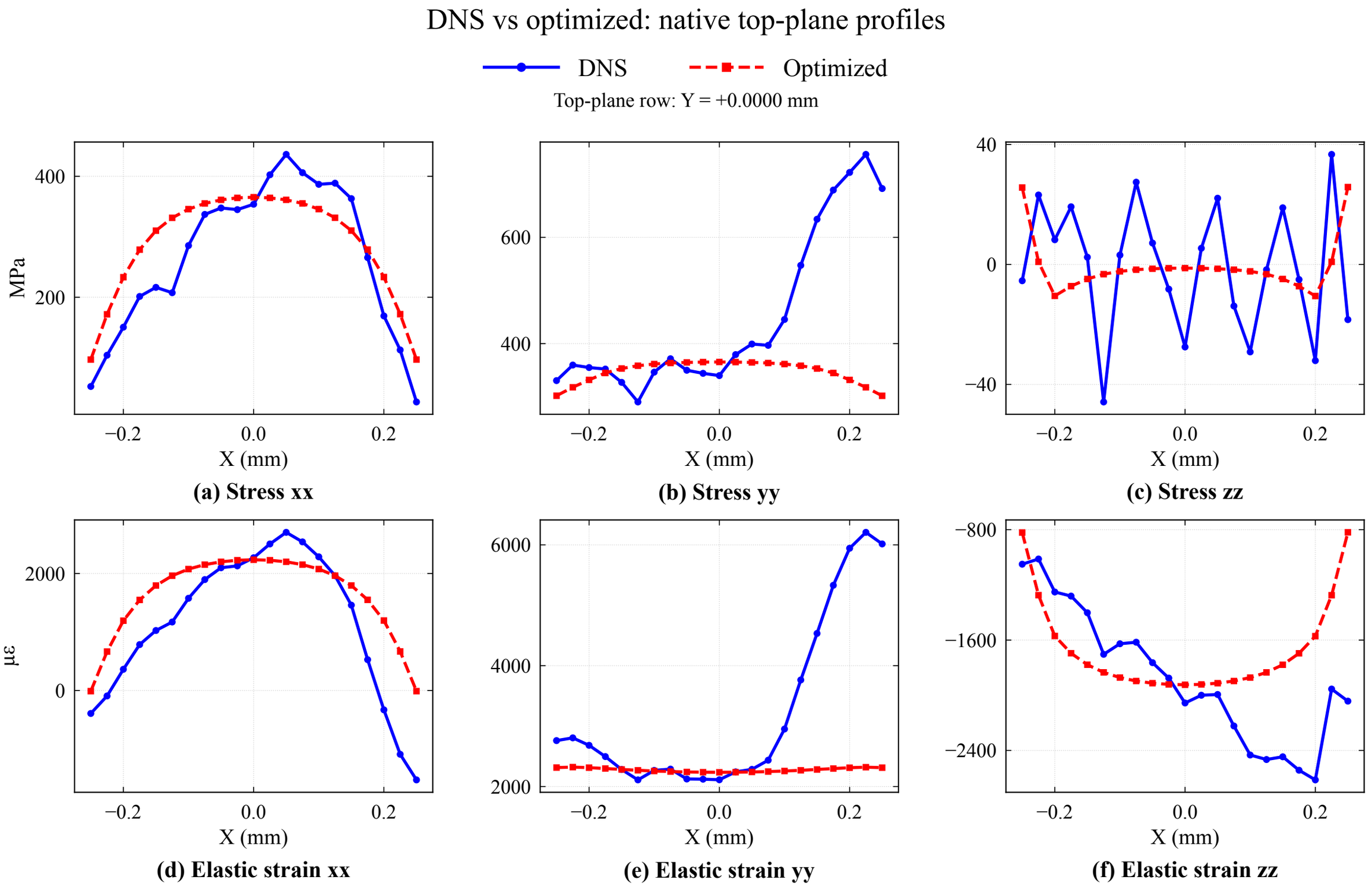


(f) Elastic strain zz



(a) DNS: Stress xx **(b) Optimized: Stress xx** **(c) Difference: Stress xx**

(d) DNS: Elastic strain xx **(e) Optimized: Elastic strain xx** **(f) Difference: Elastic strain xx**


DNS vs optimized: native top-plane profiles
DNS
Optimized
Top-plane row: Y = +0.0000 mm
MPa
400
200
−0.2
0.0
0.2
X (mm)
(a) Stress xx
600
400
X (mm)
(b) Stress yy
40
0
−40
X (mm)
(c) Stress zz
με
2000
0
X (mm)
(d) Elastic strain xx
6000
4000
2000
X (mm)
(e) Elastic strain yy
−800
−1600
−2400
X (mm)
(f) Elastic strain zz